Ontario Business Registry Web Services Addendum

Supplement to the Web Services Specification

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Date	25 July 2025
Version	1.7

Acknowledgments

The Ministry intends to be consistent with the documentation produced by other Ontario governmental projects. The content of some sections might be similar or reproduce content from other technical assessment guides.

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1 Routing By Register

To improve the stability and performance of the OBR application, areas of functionality have been segmented to a list of registers. As a result of this segmentation, messages submitted via various webservices need to be modified so they are routed to the appropriate register. To support this routing, a header which specifies the register must be included with the request. Review Table 1. Register Headers

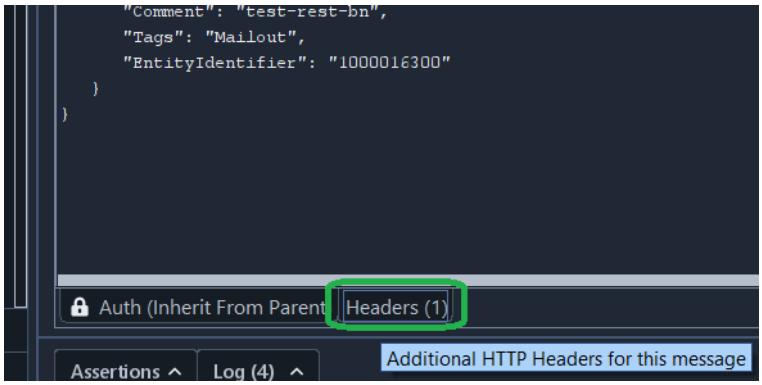
TABLE 1 – REGISTER HEADERS

Entity type	REST Header	Parameter value
Corporation	Registertype	corporationentity
Business name	Registertype	businessentity
Partnership	Registertype	partnerentity

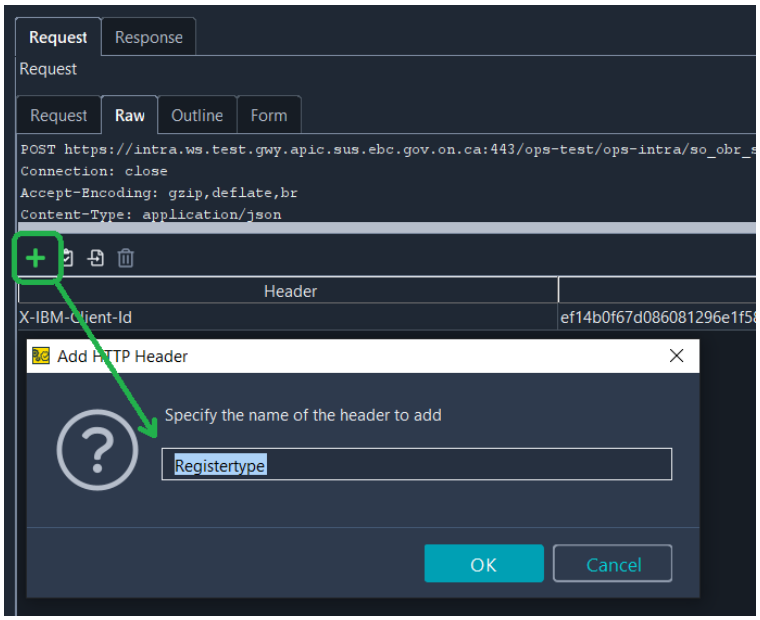
1.1 Adding the Registertype Header in ReadyAPI

1. Open the Webservice request.

2. Select **Headers**



3. Select **+** and enter *Registertype* in **Specify name of header to add** field



4. Select **OK**

5. Enter appropriate value by referring to Parameter value column in Table 1 Register Headers

6. Return to full ReadyAPI request view by selecting **Headers** again.

1.2 Webservices Using the Registertype Header

- OnbisGetAuthority
- OnbisGetEntity
- OnbisGetEntityDocumentList
- OnbisCheckEntityName
- OnbisGetEntityDocumentList
- OnbisRequestDocumentCopies
- OnbisRequestEntityProfileReport
- OnbisRetrieveProfileReport
- OnbisRetrieveServiceDocuments

The list of webservices requiring the Register Header may require some updates as web services are added or changed with later releases. You will be notified if any updates need to be made.

2 Retry Policy

For increased reliability and stability of the OBR application, Web Service users must implement the OBR Retry Policy for API calls.

Too short intervals or too many retries can have an adverse effect on the target resource or service. This may prevent the resource or service from recovering from its overloaded state, and it will continue to block or refuse requests. This results in a vicious cycle where more and more requests are sent to the resource or service, and consequently its ability to recover is further reduced.

The following Retry policy is to be implemented to be set at a **maximum of 3 retries** of the same transaction (4 attempts total). Timing: allow transaction to timeout (120s) and wait 30s (maximum of 150s between attempts).

The retry policy is applicable to ALL web services and ALL web service consumers.

2.1 Timeout Scenario

No response returned.

Attempt Number	Cycle Interval
Attempt 1	120 seconds to timeout. Wait 30 seconds.
Attempt 2	120 seconds to timeout. Wait 30 seconds.
Attempt 3	120 seconds to timeout. Wait 30 seconds.
Attempt 4	120 seconds to timeout. Fail transaction and open support ticket

2.2 Non-Timeout Scenario

Response returned but does not contain expected valid payload.

Attempt Number	Cycle Interval
Attempt 1	5 seconds to response. Unexpected/Incomplete. Wait 30 seconds.
Attempt 2	5 seconds to response. Unexpected/Incomplete. Wait 30 seconds.
Attempt 3	5 seconds to response. Unexpected/Incomplete. Wait 30 seconds.
Attempt 4	5 seconds to response. Unexpected/Incomplete. Fail transaction and open support ticket.

2.3 Document "Not Ready" Scenario

Response returned with "Not Ready" payload. Maximum number of attempts must not exceed 10.

Attempt Number	Cycle Interval
Attempt 1-9	Response received document "Not Ready". Wait 30 seconds.
Attempt 10	Response received document "Not Ready". Fail transaction and open support ticket.

3 User Authorization

User authorization for the Web Services has three requirements:

1. The client-side IP address is on the approved list.
2. The inbound message carries the appropriate certificate.
3. The inbound message carries the appropriate client ID header, which matches the certificate.

The certificate and client ID are described in more detail below.

3.1 Certificate

Requests will be rejected by the web services gateway if the request does not contain the correct certificate information. Secure access to the web services is enabled by means of standards-based public key certificates. Public key certificates using X.509 format (https://www.itu.int/rec/T-REC-X.509-201910-I/en).

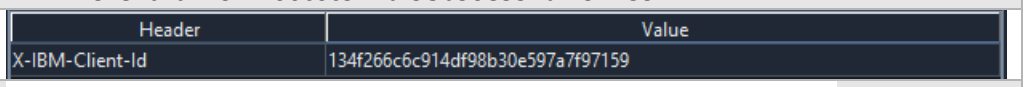
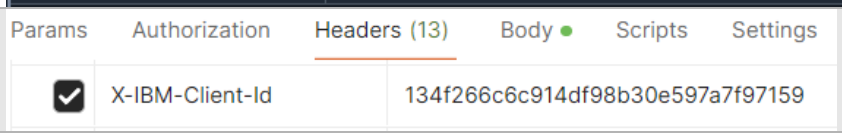
Only licensees who have worked with the Ministry to set up a valid security configuration will be able to access the web services successfully. A licensee approved to integrate with OBR web services will be provided with two X.509 certificates (one for production access and one for non-production access). Note that licensees will be required to use different certificates for production vs. non-production environments.

The Ministry will use the certificate to generate an internal security token that will identify and authenticate the web service client so that when a licensee makes any web service request that includes the certificate, the transaction will be associated with the correct OBR Organization User account.

3.2 Client ID

IBM API Connect (APIC), the API gateway that the Ministry uses for API management. APIC will automatically generate an identifier called the Client ID ('*X-IBM-Client-Id*') that uniquely identifies a new licensee. Both the certificate and the Client ID string must be present in **every** web service call from the licensee.

This Client ID must be present in the header of each web service request message. The header format should look like:

Format	Appearance
Raw (text)	X-IBM-Client-Id: 134f266c6c914df98b30e597a7f97159
ReadyAPI	
Postman	

4 Throughput Management Plan

OBR uses Throughput Management Plans (“Plans”) to control the throughput of the web services. Plans control which web services a consumer can use. Plans use two values to control the throughput: Rate limits and Burst limits.

Rate limits ensure that only a predefined maximum number of calls will be accepted within a defined time period, for example 1000 calls per hour. When the limit is reached, no more calls are accepted from that licensee until the beginning of the next time period. In this example, if a licensee makes 1000 calls in the first 20 minutes, they cannot complete any more calls until the hour has expired.

Burst limiting is similar to rate limiting but is applied to a very small time period (typically in multiples of seconds or minutes), for example 90 calls per minute. When a burst limit is reached, calls are not accepted until the enough time has passed to accommodate more API calls within the burst limit timeframe. As time progresses, the historical count of transactions over time reduces. Further web service calls continue to be accepted until the overall rate limit is reached. Burst limiting prevents usage spikes and ensures that the rate limit is evenly spread across its overall time period.

The default plan has the following limits:

- Rate limit = 3000 per 30 minutes; hard limit
- Burst limit = 200 per minute

These are current values and are subject to change as the user base and OBR technology stack changes over time.

5 Special characters Handling

For Special characters like Inverted commas (“) and Backslash (\), an escape character, also a Backslash (\) must be added before the character to pass. This will be essential when customized text is used in Shares and Provisions sections in Incorporate BCA request.

Example - \" or \\

5.1 French Accented characters and Merged symbols Handling

For any French Accented characters (ex: À, Â, Ä, Ê, É) and Merged symbols (Œ), equivalent Unicode value must be used instead of the character or symbol.

Example – Unicode value for À is \u00c0 or Unicode value for Œ is \u0152.

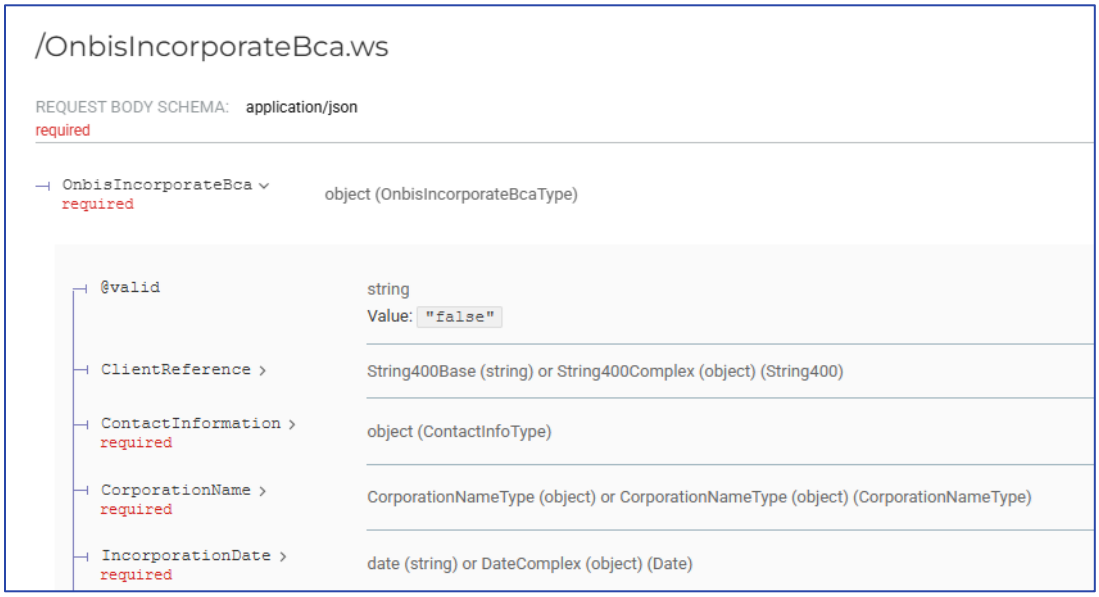
This is essential when any text field uses French Accented characters or Merged symbols.

6 JSON Element Ordering

The OBR application requires that inbound REST / JSON requests follow the same ordering as the JSON schema. We understand that it is industry standard that order is not enforced in REST / JSON. This is still true, and any inbound request with elements out of order will pass JSON validation at the gateway. However, such messages will not be parsed by the OBR application and will fail (either partially or fully).

It is required that inbound JSON requests have element ordering which conforms to the published specifications.

For an example to illustrate this point, the below specification has a defined order:



The element order is { *ClientReference, ContactInformation, CorporationName, IncorporationDate* }

If an inbound message had *IncorporationDate* before *CorporationName*, the message is valid JSON/REST, but OBR would reply with a failure. Some failures are partial, where the transaction is successful but with the offending portion dropped. Other failures are more complete, replying with an HTTP 422 response.

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